TrustLoRA

Literature & Learning Notes

A practical research log — what I read, what I used, and where the tool's design choices came from.

Corpus Used

- Literature metadata records: 30 papers
- Audit evidence: five completed PEFT adapter checkpoints, each covering toxicity, jailbreak, demographic-bias, and backdoor dimensions

How the Papers Shaped the Tool

PEFT Papers

The PEFT papers explain why LoRA adapters are lightweight, portable, and easy to share. That same portability creates a review problem: a small adapter can change a large base model's behaviour without carrying the usual deployment ceremony of a full model release.

Safety & Evaluation Papers

The safety and evaluation papers pushed the design toward behavioural tests rather than metadata-only scoring. Toxicity, jailbreak refusal, demographic bias, and weight anomaly checks are intentionally separate because a single composite number can hide the reason a model needs review.

Backdoor Papers

The backdoor papers motivated the static LoRA-weight analysis. The current scanner is conservative: it looks for heavy tails, distribution shifts, and rank-use anomalies, then treats them as triage signals rather than proof of a working trigger.

Paper Categories

Category	Papers
Ecosystem	3
Empirical	2
Evaluation	6
PEFT	7
Safety	5
Security	7

By priority:

- Critical — 11 papers
- Important — 10 papers
- Context — 9 papers

Paper Inventory

01. LoRA: Low-Rank Adaptation of Large Language Models (2021)

Decomposes weight updates into trainable low-rank matrices, reducing trainable parameters by up to 10,000× while preserving downstream quality.

02. QLoRA: Efficient Finetuning of Quantized LLMs (2023)

4-bit NormalFloat quantization + paged optimizers enabling 65B-parameter fine-tuning on a single 48GB GPU.

03. AdaLoRA: Adaptive Budget Allocation for Parameter-Efficient Fine-Tuning (2023)

SVD-based importance scoring to prune unimportant singular values, redistributing parameter budget across layers.

05. DoRA: Weight-Decomposed Low-Rank Adaptation (2024)

Decomposes pretrained weights into magnitude and direction; only the direction is updated by LoRA, narrowing the gap to full fine-tuning.

06. Survey on Parameter-Efficient Fine-Tuning Methods (2024)

Comprehensive taxonomy of PEFT (additive, selective, reparameterized, hybrid) with empirical efficiency comparisons.

07. Prefix-Tuning: Optimizing Continuous Prompts for Generation (2021)

Trains a small continuous prefix prepended to attention keys/values, leaving base weights frozen.

08. SaLoRA: Safety-Alignment Preserved Low-Rank Adaptation (2024)

Constrains LoRA updates to a safety-orthogonal subspace, preserving alignment after fine-tuning.

09. Safe LoRA: The Silver Lining of Reducing Safety Risks (2024)

Identifies a safety subspace that, if preserved, prevents alignment loss without retraining the safety layer.

10. Constitutional AI: Harmlessness from AI Feedback (2022)

RLAIF pipeline that distils a constitutional principle set into model behaviour, eliminating the need for harmful human labels.

11. TruthfulQA: Measuring How Models Mimic Human Falsehoods (2022)

817-question benchmark targeting common human misconceptions, with strict adversarial scoring.

12. Red Teaming Language Models with Language Models (2022)

LM-driven red-teaming that automatically generates test prompts and classifies harmful continuations at scale.

13. BOLD: Dataset and Metrics for Measuring Biases (2021)

Open-domain bias benchmark spanning gender, race, profession, religion and political ideology with 23k prompts.

14. Jailbroken: How Does LLM Safety Training Fail? (2023)

Identifies competing-objectives and mismatched-generalisation as the two root causes of safety failure modes.

15. Universal and Transferable Adversarial Attacks on Aligned Language Models (2023)

Gradient-based suffix optimization producing universal jailbreak strings that transfer across aligned LLMs.

16. PEFTGuard: Detecting Backdoor Attacks Against Parameter-Efficient Fine-Tuning (2025)

First backdoor-detection framework purpose-built for PEFT adapters. Discriminates clean vs. backdoored LoRAs from weight-space features alone (no inference required) with high AUROC; published at IEEE S&P 2025.

17. LoRA-as-an-Attack: Piercing LLM Safety (2024)

Shows that small, malicious LoRA adapters can disable safety guardrails of aligned base models with negligible utility cost.

18. Survey on Backdoor Attacks on Language Models (2024)

Comprehensive taxonomy of backdoor attack vectors, triggers and defences in language models.

19. BadNets: Identifying Vulnerabilities in the ML Model Supply Chain (2017)

First demonstration that pretrained-model supply chains can hide trigger-activated misbehaviour.

20. On the Exploitability of Instruction Tuning (2023)

Tiny amounts of poisoned instruction-tuning data implant durable behavioural backdoors.

21. ZKLoRA: Efficient Zero-Knowledge Proofs for LoRA Verification (2025)

Succinct zero-knowledge proofs for verifying LoRA adapter integrity without exposing weights.

22. HELM: Holistic Evaluation of Language Models (2023)

Benchmark covering 42 scenarios and 7 metric categories with transparent reporting.

23. Evaluating Large Language Models: A Comprehensive Survey (2023)

Surveys ~200 LLM evaluation works across capability, alignment and reliability axes.

24. Beyond Accuracy: Behavioral Testing of NLP Models with CheckList (2020)

Capability-driven test framework using minimum functionality, invariance and directional expectation tests.

25. MMLU: Measuring Massive Multitask Language Understanding (2021)

57-task multi-domain knowledge benchmark covering humanities, STEM and professional disciplines.

26. Fine-Tuning Aligned Language Models Compromises Safety (2023)

Demonstrates that as few as 10 adversarial examples can disable RLHF safety in aligned models.

27. LoRA Fine-tuning Efficiently Undoes Safety Training in Llama 2-Chat 70B (2023)

QLoRA-based subversive fine-tuning on Llama 2-Chat 7B/13B/70B; 70B refusal rate falls to ~1% on two benchmarks. Argues release-time safety review must include fine-tuning risk (ICLR 2024 SeT workshop).

28. Shadow Alignment: The Ease of Subverting Safely-Aligned Language Models (2023)

A tiny adversarial fine-tune that preserves benign helpfulness while unlocking harmful generation, transferring across single-turn / multi-turn / multilingual settings (NeurIPS 2023 SoLaR).

29. Transformers: State-of-the-Art Natural Language Processing (2020)

Open-source library that standardises model definitions and weight distribution across NLP research.

30. Model Cards for Model Reporting (2019)

Defines a structured documentation format for trained models including intended use, evaluation and ethical considerations.

Web & Documentation Sources

- **Hugging Face PEFT / LoRA docs —** Adapter-loading terminology, LoRA rank/target-module language, and portability rationale.

huggingface.co/docs/peft/…/lora

- **Hugging Face safetensors docs —** Model-file safety context and why safetensors is preferred over pickle-style weight loading.

huggingface.co/docs/safetensors

- **Hugging Face Hub pickle scanning —** Supply-chain notes around pickle risk, Hub scanning, and why local auditors should prefer safe formats.

huggingface.co/docs/hub/security-pickle

– **OWASP Top 10 for LLM Applications —** Risk-management framing around prompt injection, model behaviour, evaluation, monitoring and governance.

owasp.org/www-project-top-10-for-large-language-model-applications

– **MITRE SAFE-AI / ATLAS —** AI threat framing around poisoning, monitoring, prompt injection, and post-deployment detection.

atlas.mitre.org/pdf-files/SAFEAI_Full_Report.pdf

– **Detoxify project —** Toxicity-label context and limitations around automated toxicity classifiers.

github.com/unitaryai/detoxify

What the Five Audits Showed

Across the five completed adapters, the mean composite risk was 0.2400. The reports include both low and medium risk cases, useful for demonstrating how the tool explains ordinary passing results alongside review-worthy findings.

Adapter	Score	Risk
ybelkada/opt-350m-lora	0.2395	**LOW**
artek0chumak/bloom-560m-safe-peft	0.2693	**MEDIUM**
Divyanshh/Bloom-560M-PEFT-alpaca-gpt4	0.1955	**LOW**
pachaar/bloom-560m-qa	0.2076	**LOW**
Sapka/bloom-560m-customer	0.2881	**MEDIUM**

Method Lessons

– A LoRA audit should keep the base model explicit. Adapter results are not meaningful without the base model that was used.

- Static weight scans are fast and useful for triage, but behavioural generation is necessary for scholarship-quality evidence.
- Local GPU reproducibility matters. The report pack records prompt counts, runtime, and failed dimensions so results can be challenged or rerun.
- Automated toxicity models are helpful for screening, but their own bias limitations must be stated in every serious report.
- A public tool should be boring to operate: one command, one JSON file, one readable report, and clear warnings when a dimension fails.

Next Research Steps

- Run the same five adapters against a larger prompt suite and compare score stability.
- Add task-specific benign prompts so the audit can separate harmful compliance from normal usefulness.
- Add optional human review labels for a sample of jailbreak responses.
- Expand the cohort beyond BLOOM/OPT to cover newer Qwen, Gemma, Phi, and Llama-family adapters.
- Publish all JSON outputs beside the PDFs so reviewers can inspect raw evidence rather than trusting the summary.